Markov State Modelling of the HP35 Domain
Final Project for the Course "Biological Datasets for Computational Physics"

Author: Miriam Zara, MSc student in Physics of Data, University of Padova
Course Instructor: Prof.Dr. Monika Fuxreiter, University of Padova
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I. INTRODUCTION

The villin headpiece subdomain HP35 is one of the smallest proteins that folds autonomously, and one of the fastest. Its native fold is a compact three-helix bundle whose stability derives largely from a hydrophobic core formed by the aromatic residues *Phe6*, *Phe10* and *Phe17*. The Lys24-Nle/Lys29-Nle double mutant (Fig.1) studied here was experimentally found to fold in $\simeq 0.7\,\mu$s at 300 K with little temperature dependence [1]. Because of its short folding time, already in the early 2010s it has been possible to achieve long unbiased all-atom simulations that resolve dozens of folding and unfolding events [2], which made it become a paradigmatic model system for protein folding studies. A protein that folds this fast, over so low a barrier, is expected to populate only marginally stable intermediates that are difficult to resolve; yet a strict two-state description has never fully accounted for it. Both experimental [3] and simulation based studies [4–6] have reported evidence of structural heterogeneity within its conformational ensemble.

In this project, I analyse the well-known $305\,\mu$s trajectory of the Lys24-Nle/Lys29-Nle mutant published by Lindorff-Larsen *et al.* [2], which estimated the folding time at $3.2 \pm 0.6\,\mu$s [9]. I build Markov State Models (MSMs) [7] to ask whether HP35 possesses resolvable metastable substructure (e.g. off-pathway traps, intermediates, or substates within the folded basin itself). Building MSM and here a difficulty arises that is not merely technical. The MSM pipeline (feature selection, dimensionality reduction, clustering) has no unique optimal recipe, and different reasonable choices can emphasise different slow processes. Any metastable state one reports may therefore be a robust property of the protein, or an artefact of the modelling pipeline. This motivates a deliberately redundant design: rather than committing to one pipeline, I build several and ask which conclusions survive across all of them.

Concretely, four MSMs are constructed from two internal-coordinate representations (residue–residue contacts and backbone dihedral angles) each reduced by two methods, PCA and tICA. After spectral analysis and PCCA++ coarse-graining, the four models are compared on (i) the estimated folding timescale, (ii) the robustness of the resolved macrostates, and (iii) their structural interpretation.

II. METHODS

A. Dataset assembly

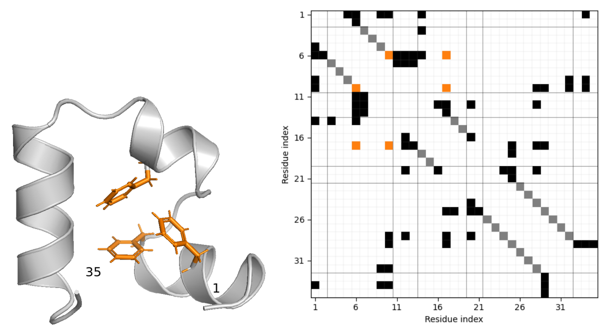

FIG. 1: ***Crystal structure and contact map of HP35*** *(PDB: 2F4K [1]). Left: The folded structure consists of three α-helices (3-10), (14-19), and (22-32), connected by two loops. Highlighted in orange are aromatic residues Phe6, Phe10, Phe17, whose interactions contribute to the formation of the hydrophobic core. Visualization with PyMOL [15]. Right: contact map computed with a distance cutoff of $4.5\,A$. Highlighted in grey are the intra-helix contacts $(i,\, i+4)$, and in orange the contacts between the three mentioned aromatic residues.*

While the original trajectory is reported by Lindorff-Larsen *et al.*[2], in this project a preprocessed version of that trajectory is employed, which is publicly distributed by Nagel *et al.* [8]. This dataset contains several timeseries of different internal coordinate representations of the protein. Each timeseries consists of approximately 1.5×10^6 samples, with a sampling interval of $\simeq 200$ ps. Authors provide timeseries of two internal-coordinate representations which they selected and filtered specifically for employment in MSM construction. These are residue-residue contact distances (specifically, `hp35.mindists2` - 42 features) and backbone dihedral angles (specifically, `hp35.dihs.shifted` - 62 features). Timeseries of residue-residue distances for a separate set of contacts identified from the crystal structure PDB 2F4K, (`hp35.crystaldists` - 53 features) are also provided, which I will use as an additional structural descriptor in the analysis. Details about the feature definition and filtering are reported in [8]. While the full all-atom timeseries was not distributed for copyright reasons (and temporarily unavailable from D.E.Shaw research due to maintenance of their databases) I was able to reconstruct the timeseries of backbone atoms using the timeseries of dihedral angles (`hp35.dihs`). All the visualizations I show in the following are realized with this trick.

B. Methods

All code is written in Python and employs the library Deeptime [10] for MSMs construction and validation. Analysis of the crystal structure (PDB 2F4K) was made using the library BioPython [11].
The construction of a MSM involves the discretization of the continuous dynamics of a high-dimensional system into a finite set of states and transitions observed at discrete time intervals. The goal is to achieve a reduced representation that maximally preserves information of relevant dynamical processes of the original system. The pipeline for MSM construction employed in this project consists of the following steps:

1. Feature selection (distances or dihedrals),
2. Dimensionality reduction (PCA or tICA),
3. Clustering of the reduced coordinates into discrete states (kmeans),
4. Estimation of the transition matrix, selection of Markov lag time τ_{MSM} (implied timescales convergence test, see description in Supp. Methods B or [7, Ch. 2.4.1])
5. Validation of the final model against the data (Chapman-Kolmogorov test, see description in [7, ch. 4.8]).

This section will mostly focus on the first two steps, which are the ones that were mainly explored in this work: feature selection and dimensionality reduction. These are used to compress the raw MD data in a lower dimensional space, so that clustering becomes feasible. Clustering algorithms typically rely on Euclidean distance, so it is important that conformations that interconvert rapidly are mapped to nearby points in the reduced space. Failure to achieve this can lead to poor markovianity [7].

Feature Selection: The two sets of internal coordinates: `hp35.mindists2` and `hp35.mindists2` are selected as features. As suggested in [8], a gaussian low-pass filter with $\sigma = 2\,ns$ is applied to the time series of each feature. This is expected to improve the MSM construction by removing high frequency recrossings between states.

Dimensionality reduction: Two classical dimensionality-reduction methods are compared: principal component analysis (PCA) and time-lagged independent component analysis (tICA). Both construct new coordinates as linear combinations of the original features. PCA identifies directions of maximal variance, whereas tICA identifies directions of maximal autocorrelation at a user-defined lag time, (τ_{tICA}). Consequently, PCA captures the largest structural fluctuations, while tICA emphasizes the slowest dynamical processes. Because Markov state models aim to resolve slow kinetics, tICA is generally preferred. Configurations that are far apart in tICA space tend to interconvert slowly, whereas rapidly interconverting configurations remain close. Thus, tICA provides a closer correspondence between geometric separation in the reduced space and kinetic separation in the underlying dynamics. (see Supplementary Methods B S1 for details). Overall, four MSMs are built, combining the two feature sets and the two dimensionality reduction methods: tica-distances, tica-dihedrals, pca-distances and pca-dihedrals. Since HP35 is known to fold at the microsecond timescale, the autocorrelation should be inspected at times shorter than this timescale. Hence, lag times between 100 and 1000 frames (approximately $20-200$ ns) were considered for τ_{tICA}.
The choice of the number of components to retain in tICA and PCA was guided by looking at their free-energy profiles and the projected time series (see Supp. Figure SF1). The rationale is that coordinates relevant to the folding process should separate in distinct metastable regions, therefore displaying multimodal free-energy profiles. Furthermore, if a coordinate captures transitions between metastable states, its time series should exhibit relatively long-lived plateaus separated by rapid transitions. Using these heuristics, 7 components for the tica-dihedrals pipeline and 5 for the other three pipelines were retained. Robustness of the first tICA components with respect to the variation of τ_{tICA} are verified and then were fixed as $\tau_{tICA} = 20\,ns$.

Clustering and selection of lag time: The number of clusters and the MSM lag time were chosen together, since the two parameters are strongly related. Increasing the number of clusters improves the resolution of the model and often makes the implied timescales converge at shorter lag times. However, it also reduces the amount of statistics available for each microstate, making the estimated transition matrix less reliable. The resulting microstate network should be strongly connected; alternatively, its largest connected component should encompass the vast majority of the trajectory frames. If the lag time is too long, or if the number of clusters is too high, some microstates may become disconnected due to insufficient sampling. The lag time should also be chosen with the process of interest in mind. Since folding in HP35 occurs on the microsecond timescale [2], the lag time should remain significantly shorter than this, otherwise intermediate kinetic details may be lost. Overall, the implied timescales test alone is not sufficient to select the lag time. The final choice was based on a compromise between implied-timescale convergence, connectivity of the microstate network, and adequate sampling of transitions. As a result, the selection remains partly qualitative. The reduced coordinates were clustered using k-means, experimenting with different numbers of clusters ($k = 30$, 50, and 100). Cluster centers were initialized using the k-means++ algorithm, and the algorithm was run for maximum 100 iterations.

Estimation of transition matrices: Transition matrices are computed with a maximum-likelihood estimator with a "sliding window" scheme [7, Chap. 4.1] while enforcing detailed balance. The largest connected component is taken as the state space of the MSM.

Spectral Analysis: The implied timescales of the MSMs and the sign patterns of the corresponding eigenvalues are analyzed. PCCA++ (Perron-Cluster Cluster Analysis) [7, Ch. 2.5.1 and 2.5.2] is employed to define macrostates, using the maximal timescale gap as a criterion for choosing the number of macrostates.

C. Statistical Analysis

CK test: performed using built-in Deeptime method and qualitatively evaluated. Model self-consistency was assessed with the Chapman–Kolmogorov (CK) test as implemented in Deeptime. The microstate MSM estimated at the selected lag $\tau_{MSM} = 50\,\mathrm{ns}$ was coarse-grained into four metastable sets with PCCA++. For a range of lag times $k\tau_{MSM}$, the coarse-grained transition probabilities predicted by the reference model (propagated to $k\tau_{MSM}$ and projected onto the four sets) were compared with those from MSMs estimated independently at each lag $k\tau_{MSM}$.

III. RESULTS

A. Features and dimensionalty reduction pipelines produce different free energy projections

The free energy surface projected onto the first $2D$ components for all four pipelines is shown in Fig. 2.

2. Clustering and selection of Markov lag time: The results of the implied timescale test are reported in Fig.3 and the corresponding populations of the largest connected component in Supp. Fig.SF2. From a grid search with $n_{clusters} = \{30, 50, 100\}$ clusters and lag times between 10 and 100 ns; $n_{clusters} = 50$ and a lag time of $50\,ns$ is selected for all four pipelines, as this seemed the best compromise between resolution, markovianity and connectivity. Importantly, the slowest timescale is approximately converged at this lag time for all four pipelines and in the correct order of magnitude (μs), which is consistent with the known folding timescale of HP35. Also it can be noticed that the implied timescales depend much more strongly on the pipeline (features+dimensionality reduction) than on the number of clusters, and the PCA-pipelines give lower timescales, which suggests that they are less accurate in approximating the underlying continuous dynamics [12]. The CK test for the pca-model is shown in Fig. SF3; the remaining pipelines behave qualitatively similarly. Predicted and estimated curves track each other closely and relax toward consistent stationary values, supporting the choice $\tau_{MSM} = 50\,\mathrm{ns}$ and the Markovian approximation so that we can consider the models validated, at least at the level needed to this benchmark. ***1. Implied Timescales Spectra:***

The timescale spectra reveal substantial differences in the slow dynamics captured by the four models (Fig. 4). Among them, the PCA-distance model exhibits the largest separation between the first and second relaxation timescales. This separation suggests a Markovian process characterized by a strong metastable structure, in which the state space is effectively partitioned into two well-defined metastable sets. Within each set, probability redistribution occurs on relatively fast timescales, whereas transitions between the two sets are substantially slower. Consequently, the dominant dynamical process is associated with the rare interconversion between these metastable regions. For this model the two Perron clusters identified by PCCA++ are expected to have a straightforward structural interpretation as folded and unfolded metastable states. Under this hypothesis, the folded basin should be characterized by a narrow distribution of contact distances around the corresponding crystal values in the "folded" basin.

In the other three models, the separation between the slowest timescale and the remaining processes is less pronounced. The tICA-dihedral model departs most strongly from a two-state description, exhibiting a slowly decaying timescale spectrum. Although its dominant timescale is comparable to those of the distance-based models, several additional timescales remain above the resolution threshold, suggesting a more complex metastable structure than a simple two-state decomposition.

As a first step towards model comparison we construct a two-state PCCA++ decomposition for each of them, including those whose timescale spectra suggest more complex dynamics. We then examine the extent to which the resulting metastable sets agree. Since the dominant implied timescale is similar across three of the models, it is natural to ask whether it corresponds to the folding–unfolding transition. A first indication that the slowest dynamical process corresponds to folding and unfolding

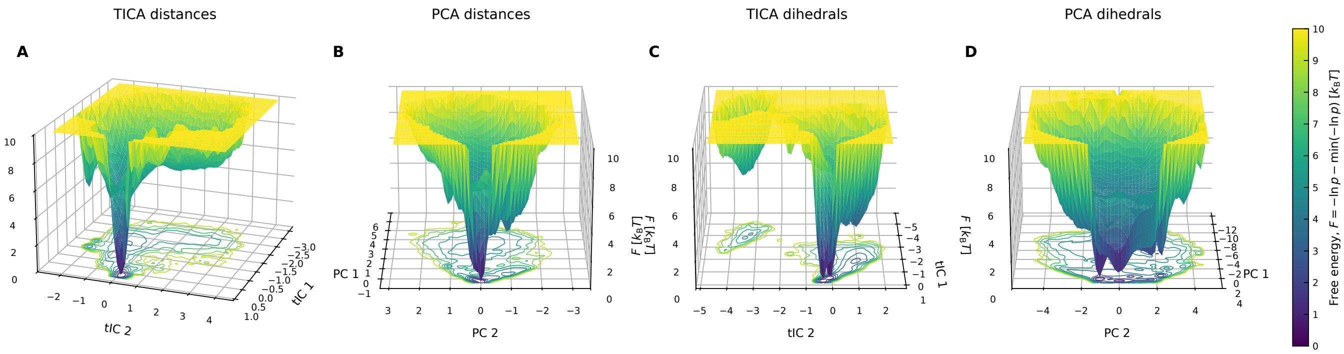

FIG. 2: ***Two-dimensional free-energy surfaces projected onto the first two reduced coordinates.*** *Because projection can obscure barriers and metastable states, these landscapes should be interpreted qualitatively. Nevertheless, substantial differences are observed among the four pipelines. Distance-based models exhibit a single dominant minimum, whereas dihedral-based models show two major minima. The tICA projections additionally contain a low-free-energy basin well separated from the main funnel.*

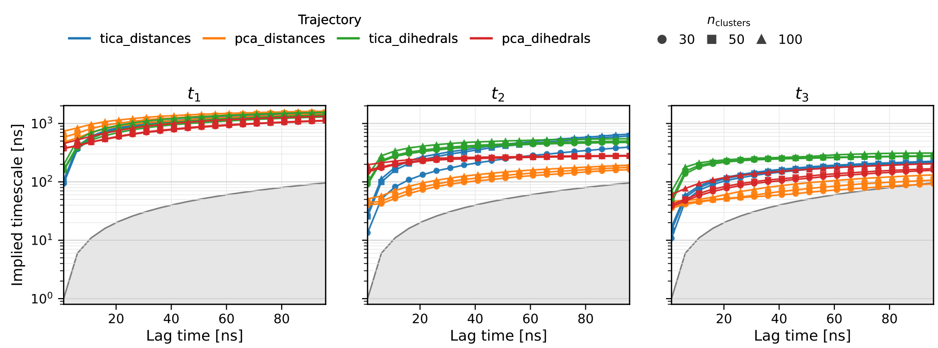

FIG. 3: ***Results of the timescale convergence test****: First three implied timescales as a function of the MSM lagtime. Colors denote the four combinations of feature set and dimensionality reduction method, while marker styles indicate the number of microstates used in the state-space discretization. The shaded region corresponds to $t_i \leq \tau_{MSM}$, where t_i is the implied timescale and τ_{MSM} the MSM lagtime. Timescales falling within this region are not resolved by the model, as the associated relaxation process occurs on a timescale shorter than a single Markov-chain transition step.*

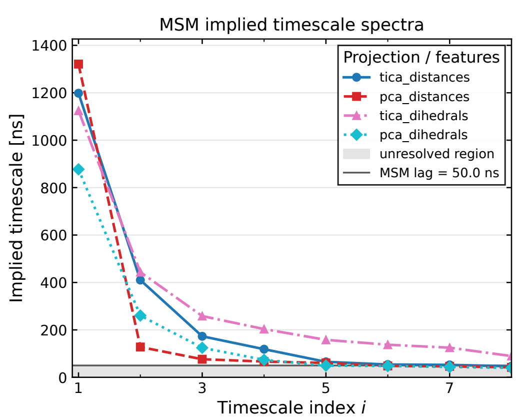

FIG. 4: ***Relaxation timescales spectra*** *at selected lag time $\tau_{MSM} = 50\,\text{ns}$. The horizontal grey line marks the MSM resolution limit ($t = \tau_{MSM}$).*

is obtained from the first right eigenvector. Its sign pattern closely follows the distribution of the fraction of native contacts, Q (Supplementary Figure SF4), suggesting that the dominant relaxation mode separates folded and unfolded conformations. To investigate this, we use the fuzzy state memberships provided by PCCA++ to compute membership-weighted ensemble averages, reducing the influence of configurations that cannot be clearly assigned to a single state. The fuzzy PCCA++ membership probabilities of i) microstates and ii) single trajectory frames is computed. Then, the distribution of the fuzzy membership probabilities of frames across models is checked (see Supp.Fig. SF5).The highest concordance is observed between the tICA-distance and PCA-distance models, which disagree on only $\simeq 1.6\%$ of the frames, followed by the PCA-distance and PCA-dihedral models, with a disagreement of $\simeq 2.3\%$ In contrast, the tICA-dihedral model exhibits the lowest agreement with the other representations, differing on approximately 8% of the frames.

Inspection of the PCCA++ membership probabilities for state $S1$ reveals differences among the models (Supp. Fig.SF5). The PCA-based representations produce sharply bimodal distributions, whereas the tICA-dihedral model yields substantially broader distributions and, consequently, much fuzzier assignments. This behavior is consistent with its implied timescale spectrum, which lacks a pronounced separation between the dominant processes and therefore does not naturally support a two-state decomposition. Interestingly, both dihedral-based models display a bimodal distribution of assignment probabilities, with peaks near 0.8 and 1.0 This is consistent with the free-energy projections onto the first $2D$ coordinates, which suggests the presence of two minima that might correspond to alternative folded conformations.

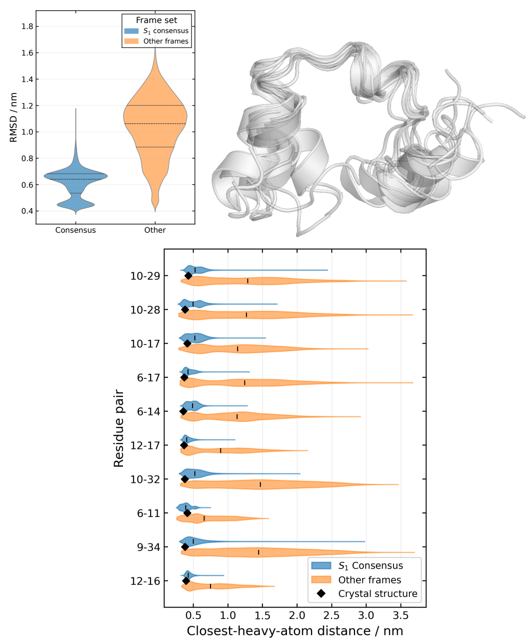

FIG. 5: **Structural Analysis of S_1 consensus frames:** *Top left: distribution of RMSD from the crystal structure. Top right: overlay of a random ensemble of* 8 *consensus frames. Bottom: distribution of most important contacts as identified by the score defined in the text.*

2. Structural caracterization of "consensus" microstates:

Consensus microstates were defined as those assigned a probability greater than 0.80 of belonging to state S_1 by all four models. This subset comprises approximately $\simeq 61\%$ of the trajectory frames. To validate the interpretation of macrostate S_1 as the folded basin, a structural characterization of the consensus frames was performed. A simple characterization of S_1 can be obtained from native contacts. For each contact observed in the crystal structure (`hp.crystaldist`), the distance distribution computed over the consensus frames was compared with the distance distribution computed from all remaining trajectory frames. Contacts that distinguish the two ensembles were identified using a robust standardized difference between their median distances:

$$\text{score}_{i,j} = \frac{\text{median}(r_{ij}^{\text{other}}) - \text{median}(r_{ij}^{\text{consensus}})}{\left[IQR_{ij}^{\text{other}} + IQR_{ij}^{consensus}\right]/2},$$

The score is expected to be positive, as the contact distances should be shorter in the "consensus" set if that represents the folded basin. A large positive score indicates that contact (i,j) is shorter in the consensus ensemble, relative to the variability of its distance distribution. The distribution for the 10 most important contacts according to this score, together with the visualization of a random subset of structures and the distribution of the Root Mean Square Deviation (RMSD) in the two sets are reported in Figure 5. From the contact analysis, the most important features for identifying the folded basin lie within the hydrophobic core, particularly those involving residues near Phe6 and Phe10. In contrast, contacts defining the three helices contribute little to the discrimination. This is consistent with the observation that substantial helical structure is present in both folded and unfolded conformations (Supplementary Figure SF6), whereas formation of the hydrophobic core is largely restricted to the folded basin. Interestingly, several observables, including the

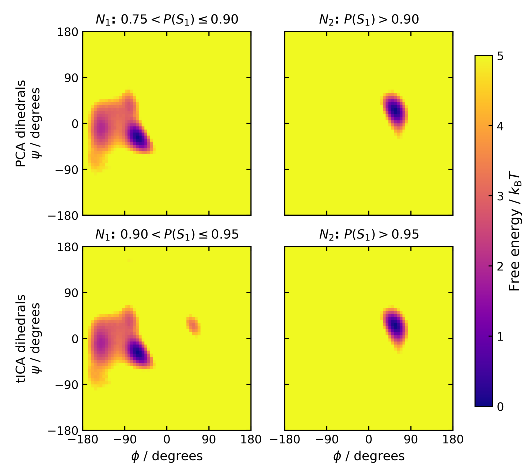

FIG. 6: ***Structural distinction between the two folded substates identified by the dihedral-based MSMs:*** *(ϕ, ψ) distribution of the residue Asp3 in the two subpopulations corresponding to the distinct minima observed within the folded basin.*

RMSD and specific long-range contacts such as 10–29, display bimodal distributions within the consensus folded ensemble. The same bimodality is visible in the free-energy projections obtained from the dihedral-based models and is reflected in the distribution of the PCCA++ membership probability for macrostate 1. These observations suggest that the folded basin identified by the dihedral-based MSMs is itself heterogeneous and may contain two structurally distinct substates. To investigate this possibility, I partition the folded ensemble identified by the dihedral-based models using the peaks in the PCCA++ membership probabilities (see Supp. Figure SF5) and compare their structural characteristics. The two sets of microstates are found to be distinguished primarily by the conformational state of a single residue, *Asp3*. In one state, *Asp3* predominantly occupies the α_R region of Ramachandran space, whereas in the other it preferentially populates the α_L region (Fig. 6).

3. Estimation of relative occupancy and residence times: Interpreting S_0 and S_1 as the unfolded and folded basins, respectively, allows a rough estimate of the folding kinetics to be obtained. Although the resulting times should be regarded as indicative only, they provide a useful basis for comparing the different models. A more rigorous analysis would require computation of mean first-passage times and reactive fluxes on the full microstate MSM using transition path theory, which lies beyond the scope of the present work. To reduce the effect of rapid recrossings between the two macrostates, dynamical coring was applied to the trajectory [13]. Frames were assigned to macrostates using a hard assignment, and a transition was counted only if the trajectory remained in the new macrostate for at least one MSM lag time ($\tau_{\mathrm{MSM}} = 50\,\mathrm{ns}$). The resulting cored trajectory is shown in Supp. Fig. SF7. The folding time, t_{fold}, is then approximated as the average waiting time between successive unfolded-to-folded transitions in this cored trajectory. Results of folding times and relative populations are reported in Fig. 7.

The results are model dependent. Overall, trajectory-based estimates of the ΔG agree within one standard deviation with the estimate of $-0.6 \pm 0.2\, kcal/\, mol^{-1}$ reported by [9] while all models predict a shorter folding time (as the mean/median residence time in the state S_0) when compared to the reference value of $3.2 \pm 0.6\,\mu\mathrm{s}$ reported in [9]. The systematic underestimation is most likely attributable to the relatively short Markov lag time used here, at which the slowest timescale has not yet fully converged. Note that in principle another estimate can be derived directly from the PCCA coarse-grained transition matrix and fuzzy membership probabilities (see Suppl. Table I).

3. Beyond two-states: three-state PCCA++ on tICA models:

The implied timescale spectra of the tICA-based models (Fig. 4) suggest the presence of additional slow dynamical processes whose characteristic timescales are comparable to that of the dominant process. For both tICA models, the largest spectral gap occurs between the second and third implied timescales rather than between the first and second. This observation indicates that a three-state kinetic decomposition may provide a more natural description of the underlying dynamics than a simple two-state partition. Consequently, a three-state PCCA++ decomposition was applied to the tICA models, and the structural characteristics of the additional metastable state were investigated.

tICA-distances: A three-state PCCA partitioning produces a lowly populated ($\sim 1\%$) additional macrostate. Key structural descriptors that distinguish this macrostate from the others are the dihedral angles of residues $Phe6, Lys7, Gly15$ which populate the β-region of the Ramachandran plot in $S0$, while they populate both the α and the β region in $S1$ and only the α_R region in $S2$ (Fig.8). A visual inspection shows this is a well defined state, caracterized by helix 3 which never unfolds, compact structure and a stable beta-hairpin formed in the region of residues $6 - 14$. At all effects this appears to be a kinetic trap in the folding process. Populations also agree, with $\simeq 1\%$ in this work and $\simeq 1.49\%$ in [4]. Inspection of the dynamically cored macrostate trajectory shows that this state is visited only once, where it remains occupied for approximately $1\,\mu\mathrm{s}$.

tICA-dihedrals: A clear structural interpretation of macrostate S_0 is not immediately apparent. Comparison of the dihedral-angle distributions (Supp. Fig. SF8) indicates that the most pronounced difference between S_0 and the other two macrostates is localized in the $Leu20$–$Pro21$ loop connecting helices 2 and 3 in the native structure. In S_0, the ϕ/ψ angles of $Leu20$ predominantly occupy the β region of the Ramachandran plot, whereas in S_1 ($\simeq$ the unfolded basin) and S_2 ($\simeq$ the folded basin) they are concentrated in the α_R region. The transition $S_0 \leftrightarrow S_1$ therefore appears to correspond to a local conformational rearrangement of the second loop within the unfolded ensemble. Inspection of the dynamically cored macrostate trajectory shows that S_0 is visited multiple times, always from S_1. The state has a stationary population of approximately 3% and a median residence time of $0.32\,\mu\mathrm{s}$.

IV. DISCUSSION AND CONCLUSIONS

This project aimed to investigate the robustness of the traditional Markov State Modelling pipeline, using as a benchmark a simple protein that has been extensively simulated and used as a model system. In particular, the central question was the investigation of the effects of i) selecting different internal coordinate representations as features ii) changing the dimensionality reduction method from PCA to tICA.

This study demonstrates that, even for a small protein, constructing a robust MSM is challenging. Although the four modeling pipelines converge on the broad features of the kinetics, they differ in their description of the finer dynamical structure. All four pipelines correctly locate the slowest relaxation process on the microsecond timescale (Fig. 4). With PCCA++ algorithm applied, they also consistently resolve a two-state separation whose native-like basin is largely model independent, with a maximum disagreement of 8% in frame assignments (Supp.Fig. SF5). The distinction between the two basins is driven primarily by a small set of tertiary contacts surrounding Phe6 and Phe10—the hydrophobic core—whereas the helical segments remain populated in both states (Fig 7). On the thermodynamic side, the resulting folded/unfolded free-energy differences bracket the value of -0.6 ± 0.2 kcal mol^{-1} reported by Piana et al. [9] and agree with it within two standard deviations across all four models. The kinetic comparison is less clean: the mean residence times in the folded basin ($\approx$1.5–5 μs from the coarse-grained matrix, $\approx$1.6–2.7 μs empirically) are of the right order but are systematically shorter than the directly measured folding time of 3.2 ± 0.6 μs reported by Piana et al. [9]. This effect is particularly pronounced for the dihedral-based models. Repeating the estimate at a longer lag (200 ns) yields estimates consistent with the reference value of Piana et al, at the cost of temporal resolution. Differences arise already in the estimated folding times and extend to the structure of the free-energy landscape and the associated Markov network. This is unsurprising, as low-dimensional projections of the free-energy surface are inherently incomplete and somewhat arbitrary. Rather than being viewed as definitive descriptions, they are best interpreted as complementary representations of the underlying dynamics. Differences arise both from changing features and from changing dimensionality reduction method. In dihedrals representation, opposed to distance

Model	$N_{\rm visits}$	Mean (μs)	Median (μs)	$P(S_0)$	$P(S_1)$	ΔG_{1-0} (kcal mol^{-1})
tICA distances	35	2.712	1.725	0.311	0.689	-0.471
PCA distances	37	2.665	1.703	0.323	0.677	-0.438
tICA dihedrals	46	1.650	1.014	0.249	0.751	-0.655
PCA dihedrals	42	2.287	1.192	0.315	0.685	-0.461

FIG. 7: *Empirical unfolded-state residence times computed from the cored macrostate trajectories, where $N_{\rm visits}$ is the number of unfolded visits observed in each trajectory. Relative free energies were calculated as $\Delta G_{1-0} = -k_{\rm B}T\ln[P(S_1)/P(S_0)]$ at $T = 360$ K.*

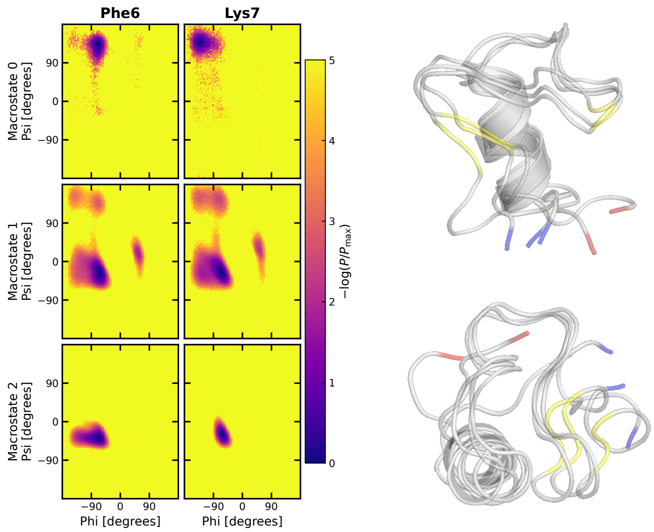

FIG. 8: ***Structural characterization of macrostate S_0 in the three-state tICA-distance model.*** *Ramachandran distributions of residues Phe6 and Lys7 together with representative conformations from macrostate S_0. This state is characterized by the formation of helix 3, while the N-terminal region adopts a β-hairpin-like structure in place of helix 1. Residues highlighted in yellow correspond to those shown in the Ramachandran plots. The N-terminus is highlighted in blue, the C-terminus in red. Visualization with Pymol [15].*

representation, additional structure is revealed as two distinct conformational substates emerge, distinguished primarily by the orientation of Asp3 and the associated structural details of helix 1 (Fig. 6). The existence of these two configurations is in agreement with the finding of both [4] and [8]. In particular, the α_L-state appears to coincide with the "$N1$, $N2$" states in [4], whereas the α_R state closely resembles their "I_1" state. Conversely, the tICA-distance model is the only representation that appears to resolve a well-defined misfolded state, characterized by a stable helix 3 and the formation of a β-hairpin in place of helix 1 (Fig. 8). Notably, this state bears a strong resemblance to the I_3intermediate identified by Stock *et al.* [4], which is likewise a compact conformation featuring a fully formed helix 3 and a β-hairpin-like structure in the region occupied by helix 1 in the native state. Jain and Stock [4], using most-probable-path clustering, described the HP35 landscape as hierarchical and identified a sparsely populated compact intermediate, I_3, in which helix 3 is formed while the N-terminal region collapses into a helix-1-like hairpin. The third state resolved here by the tICA-distances model reproduces that object closely—a compact, singly-visited trap with helix 3 intact, a β-hairpin around residues 6–14, and a population of $\simeq 1\%$ against their $\simeq 1.5\%$—so on this point the analysis in this work *supports* their hierarchical description and, *in addition*, shows that the intermediate is recoverable by a standard tICA/PCCA++ pipeline provided the features are contacts rather than dihedrals. These results also highlight that slow dynamics are not necessarily synonymous with functionally relevant or mechanistically important dynamics. In particular, the tICA-dihedral model identifies an additional slow process associated with a conformational rearrangement in the second loop. The significance of this transition within the folding mechanism remains unclear. It may represent a genuine intermediate step in the folding landscape, but its role is difficult to assess from the present analysis. Furthermore, the structural interpretation is limited by the use of a reconstructed backbone representation; access to the full atomistic coordinates could provide additional insight into the nature and potential relevance of this state. Also to be honest, in the tica-dihedrals case, where a clear eigenvalue gap is lacking, stopping at three clusters is also an arbitrary choice, instead it would be more appropriate to change the algorithm altogetehr, moving away from PCCA++ which is known for not being adeguate for such cases.

Numerous avenues remain for improving and extending the present analysis. Among the most important is assessing the robustness of the results with respect to the metastable-state identification procedure itself, for example by comparing PCCA++ with alternative approaches such as MPP, which was employed in several of the reference studies. Likewise, a more complete kinetic characterization would require construction of the full microstate MSM and a subsequent transition path theory (TPT) analysis. Despite these limitations, the project achieved its primary objective as a didactic exploration of the MSM framework. More broadly, it illustrates the challenges involved in constructing and interpreting Markov state models, even for a relatively small protein and a high-quality simulation dataset. The analysis shows that different but seemingly reasonable modeling choices can lead to distinct descriptions of the slow dynamics and metastable structure. These observations underscore the importance of rigorous validation procedures, sensitivity analyses, and standardized protocols in improving the reproducibility of MSM-based studies and, ultimately, their usefulness as tools for understanding molecular dynamics simulations.

V. REFERENCES

Data availability statement: Preprocessed trajectories of the HP35 villin headpiece are kindly made available by Nagel et al. and downloadable at their GitHub https://github.com/moldyn/HP35-DESRES. All code necessary to replicate this analysis is available on GitHub at https://github.com/miriamzara/MSM.

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Appendix A: Supplementary Figures

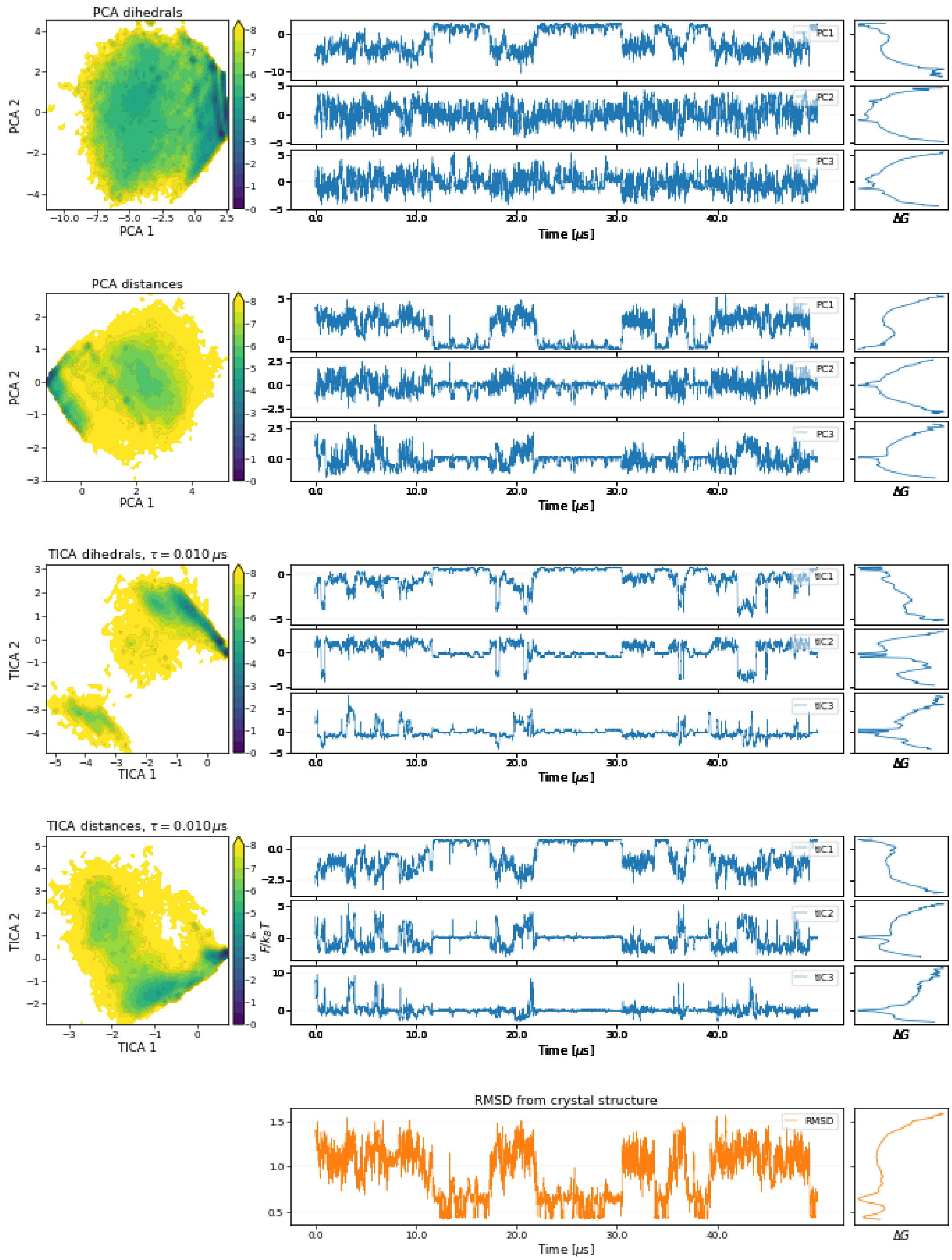

FIG. SF1: **Reduced-coordinate representations obtained from the four feature–dimensionality-reduction pipelines.** Rows correspond to PCA dihedrals, PCA contact distances, tICA dihedrals, and tICA contact distances. For each pipeline, the left panel shows the projection of the trajectory onto the first two reduced coordinates. The remaining panels show the time evolution of the first three reduced coordinates during the first $\sim 40\,\mu$s of simulation together with the corresponding one-dimensional free-energy profiles computed from the full trajectory. The bottom row reports the RMSD from the crystal structure and its free-energy profile for comparison.

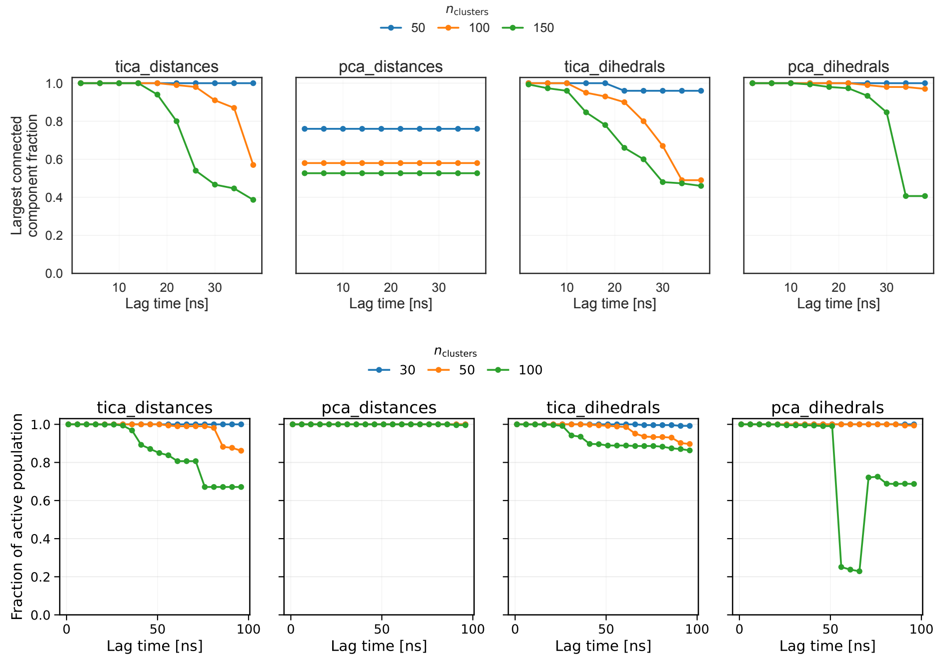

FIG. SF2: **MSM connectivity and active population as a function of lag time.** Top panel: fraction of microstates in the largest connected component of the MSM as a function of lag time. Bottom panel: fraction of the total equilibrium population contained in the largest connected component of the MSM as a function of lag time

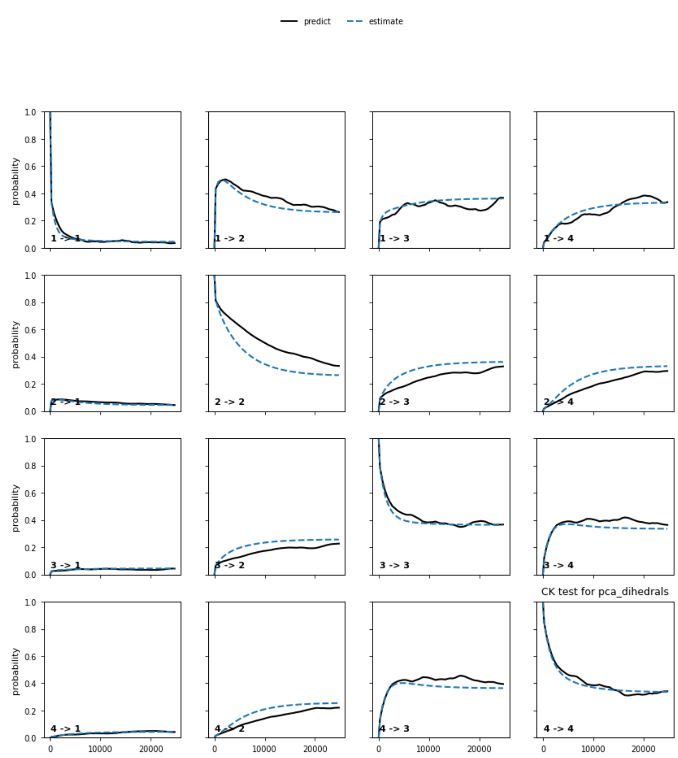

FIG. SF3: **Chapman–Kolmogorov test for the pca-dihedrals MSM**. The microstate model estimated at $\tau = 50\,\mathrm{ns}$ was lumped into four metastable sets with PCCA++. Each panel $(i \to j)$ shows the probability of being in set j at time $k\tau$ given the system started in set i. Solid lines: MSMs estimated independently at each lag; dashed lines: predictions from the $\tau = 50\,\mathrm{ns}$ model. Close agreement indicates approximate Markovianity at the chosen lag.

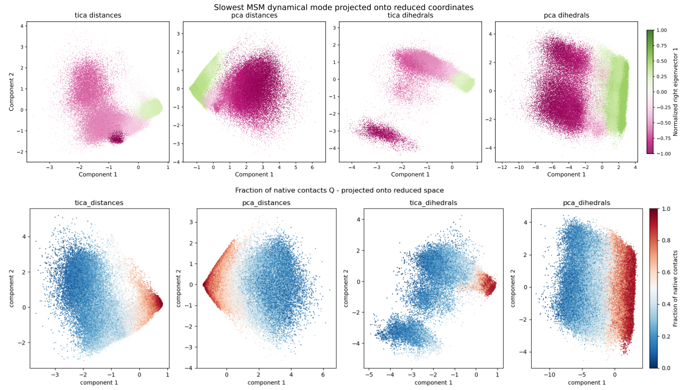

FIG. SF4: **Comparison between the slowest MSM relaxation mode and a structural folding descriptor.** Top row: values of the first non-trivial eigenvector projected onto the first two components of the reduced coordinate space. Positive and negative components identify regions that exchange probability on the slowest resolved timescale of the Markov model. Bottom row: projection of the fraction of native contacts, Q, onto the same coordinates. The close correspondence between the eigenvector sign pattern and the distribution of Q indicates that the slowest kinetic process captured by the MSM is associated with the folding-unfolding transition.

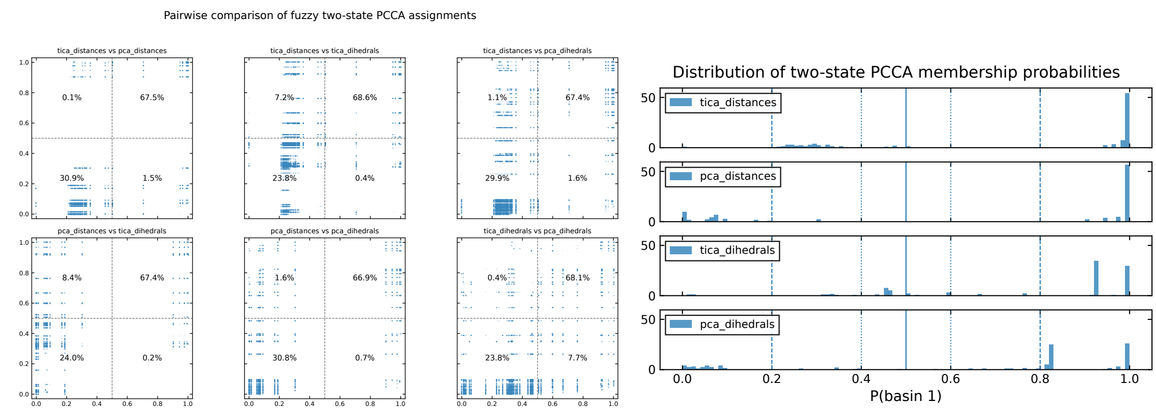

FIG. SF5: **Fuzzy membership probabilities of the trajectory frames to the two macrostates across models.** Left: Each panel shows the fuzzy membership probability of each frame in the trajectory to macrostate S_0 ($\simeq$ denatured basin) and macrostate S_1 ($\simeq$folded basin) for a different model. The models based on distances (tica-distances and pca-distances) show very similar distributions, while the tica-dihedrals model exhibits a broader distribution, particularly for frames assigned to the denatured basin. This indicates that the tica-dihedrals model has more uncertainty in assigning frames to the denatured basin. Right: Distribution of the fuzzy membership probability of each frame to macrostate 1 for the four MSMs. Probabilities close to 0 or 1 correspond to frames that are confidently assigned to a single macrostate.

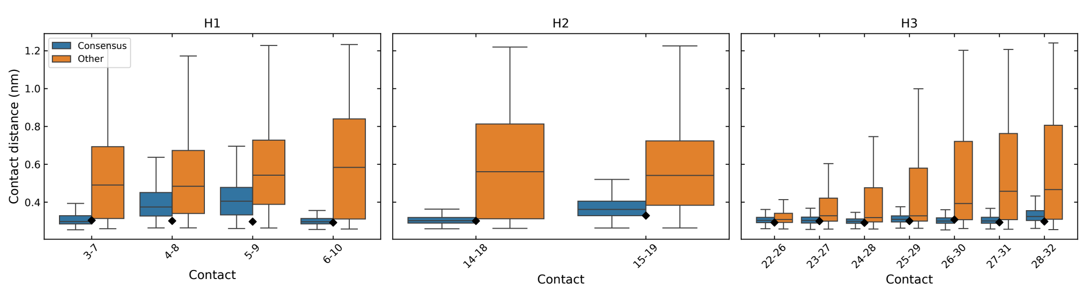

FIG. SF6: **Helix content in the consensus S_1 frames:** Distribution of distances for all intra-helical contacts in the consensus folded ensemble (S_1, blue) and in all other frames (orange). The black squares marks the distance observed in the crystal structure PDB 2F4K. Although helical contacts are generally more compact within S_1, their distributions overlap substantially with those of the remaining frames, indicating that significant helical structure is present both inside and outside the folded basin.

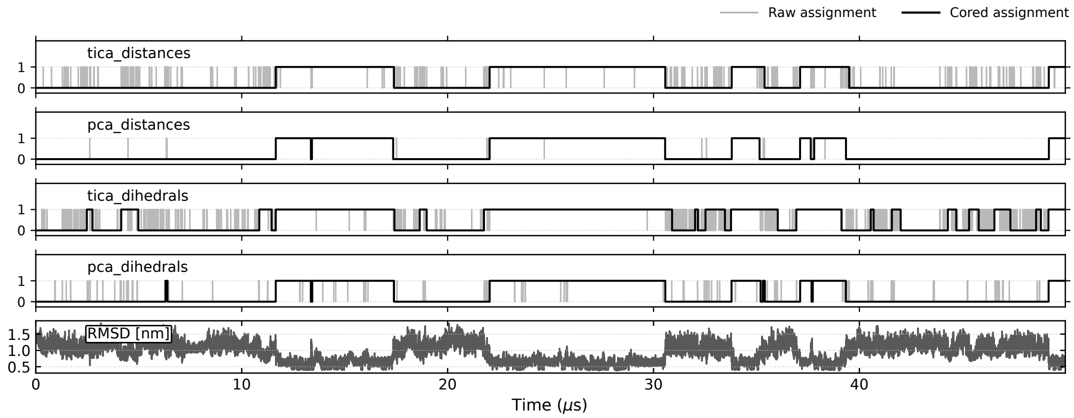

FIG. SF7: **Comparison of raw and filtered macrostate trajectories for the two-state PCCA++ models.** For each MSM, the grey trace shows the instantaneous hard assignment of each frame, while the black trace shows the filtered assignment obtained by removing transitions that persist for less than one Markov lag time ($\tau = 50\,\mathrm{ns}$). The RMSD time series is shown at the bottom for reference. This filtering suppresses rapid recrossings near the state boundary and highlights the long-lived folded and unfolded episodes underlying the dynamics. Only the first $\simeq 50\,\mu s$ of the trajectory are shown.

Model	$P(S_0)$	$P(S_1)$	$\langle t_0 \rangle$ (μs)	$\langle t_1 \rangle$ (μs)	ΔG_{1-0} (kcal mol^{-1})
tICA distances	0.230	0.770	1.591	5.312	-0.863
PCA distances	0.303	0.697	1.933	4.446	-0.596
tICA dihedrals	0.218	0.782	1.471	5.286	-0.915
PCA dihedrals	0.358	0.642	1.407	2.520	-0.417

TABLE I: **Thermodynamic and kinetic properties estimated from two-state Markov Models:** Stationary populations, mean residence times, and relative free energies obtained from the two-state Markov model induced by the PCCA++ coarse graining of each MSM. Mean residence times were computed as $\langle t_i \rangle = \tau/(1 - T_{ii})$, where T denotes the coarse-grained transition matrix and τ the MSM lag time. Relative free energies were calculated as $\Delta G_{1-0} = -k_{\mathrm{B}} T \ln[P(S_1)/P(S_0)]$ at $T = 360$ K.

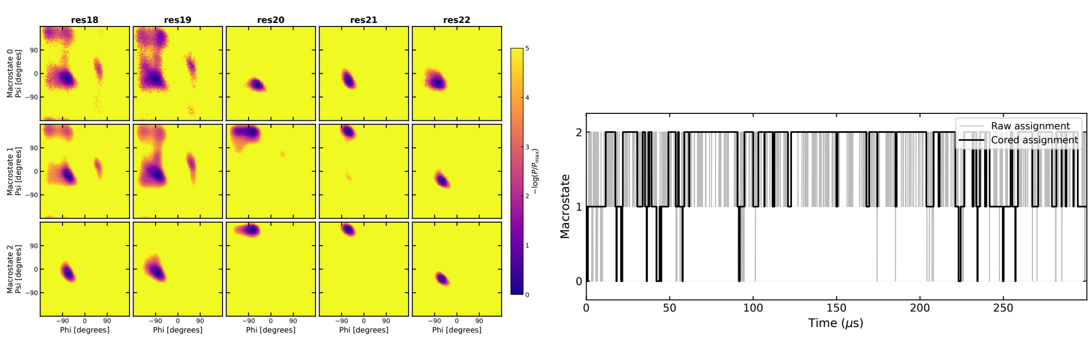

FIG. SF8: **Structural caracterization of the 3-state PCCA macrostates for the tica-dihedrals model:** Top: (ϕ, ψ) distribution of the residues $18 - 22$ in the three macrostates S_0, S_1, S_2. Bottom: trajectory of the three-state PCCA++ model for tICA-dihedrals, with hard assignations and cored assignations (transitions are counted if they remain in the destination for at least one markov step, which is $\tau = 50\,ns$).

Appendix B: Supplementary Methods

Here I provide the technical details of the procedure summarized in Methods.

S1. Derivation of PCA and tICA

The presentation of tICA in this section follows closely that of [14].

PCA and tICA are both linear dimensionality reduction techniques, i.e. they project the data onto a linear subspace of the original feature space. Let $\{\mathbf{x}(t)\}_{t=0}^{T-1}$ be a multidimensional, discrete time-series, where each snapshot is a column vector, of dimension D, corresponding to an arbitrary, vectorized representation of a protein conformation (e.g. a set of dihedral angles). Suppose the data has been sampled from a *stationary* and *time-reversible* stochastic process. Without loss of generality, I assume that the data have been centered, $\mathbb{E}[\mathbf{x}] = 0$. Using the braket notation $\langle x| := \mathbf{x}^T, |x\rangle := \mathbf{x}$, I define the covariance matrix:

$$\mathbf{C}_0 := \mathbb{E}[|x\rangle \langle x|] \tag{B1}$$

and the time-lagged covariance matrix:

$$\mathbf{C}_{0,\tau} : = \mathbb{E}\left[|x(t)\rangle \langle x(t+\tau)|\right]. \tag{B2}$$

For a stationary reversible process,

$$\mathbf{C}_{0,\tau} = \mathbf{C}_{\tau,0}^T = \mathbf{C}_{\tau,0},$$

so that the lagged covariance matrix may be denoted simply by $\mathbf{C}_\tau$.

If $|\hat{u}\rangle \langle \hat{u}|$ is a projection operator (i.e. $|\hat{u}\rangle$ is a unit vector), then:

$$\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle = \mathbb{E}[\langle \hat{u}|x\rangle \langle x|\hat{u}\rangle] = \mathbb{E}[\langle \hat{u}|x\rangle^2] \tag{B3}$$

is the variance of the projection of the data along $|\hat{u}\rangle$, and

$$\begin{aligned} \mathrm{ACF}_{\hat{u}}(\tau) &= \frac{\langle \hat{u}| \mathbf{C}_\tau |\hat{u}\rangle}{\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle} \\ &= \frac{\mathbb{E}[\langle \hat{u}|x(t)\rangle \langle \hat{u}|x(t+\tau)\rangle]}{\mathbb{E}[\langle \hat{u}|x\rangle^2]} \end{aligned} \tag{B4}$$

its autocorrelation at lag time τ.

PCA seeks the direction of maximal variance. The first principal component is defined as

$$\begin{aligned} |\hat{u}_1\rangle = &\arg\max \langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle \\ &\text{subject to } \|\hat{u}\| = 1 \end{aligned} \tag{B5}$$

this optimization problem can be solved introducing a Lagrange multiplier λ, and the stationarity condition yields

$$\mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i |\hat{u}_i\rangle, \tag{B6}$$

i.e. the projection versor is an eigenvector of the covariance matrix. The corresponding eigenvalue equals the variance of the data projected onto the principal component:

$$\langle \hat{u}_i| \mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i \cdot \langle \hat{u}_i|\hat{u}_i\rangle = \lambda_i.$$

Then the second principal component is defined as the direction of maximal variance orthogonal to the first, giving the optimization problem:

$$\begin{aligned} |\hat{u}_2\rangle = &\mathrm{argmax} \langle u| \mathbf{C}_0 |u\rangle \\ &\text{subject to } \langle u|u\rangle = 1 \\ &\text{and } \langle u|\hat{u}_1\rangle = 0 \end{aligned} \tag{B7}$$

which again can be solved with Lagrange multipliers. Iterating, this implies that the principal components form an orthonormal basis of eigenvectors of the covariance matrix, i.e. they diagonalize the covariance matrix. Dimensionality reduction ($d < D$) is achieved by taking the components along the first d principal components $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$z_i = \langle \hat{u}_i|x\rangle, \qquad \mathbf{z} = U^T\mathbf{x}. \tag{B8}$$

This is a linear projection onto an ortonormal basis:

$$|x\rangle \to |z\rangle := \sum_{i=1}^{d} \langle x|\hat{u}_i\rangle \, |\hat{u}_i\rangle$$

By construction, in the reduced space components are uncorrelated and ordered by decreasing variance:

$$\mathbb{E}[|z\rangle \langle z|] = \mathrm{diag}(\lambda_1, \cdots \lambda_d) \tag{B9}$$

tICA Unlike PCA, tICA [14] seeks directions $|u\rangle$ that maximize the autocorrelation Eq.(B4) at a fixed lag time τ. The lagtime τ is a free parameter. The autocorrelation is invariant under scaling of the projection vector $|u\rangle$. Differently from PCA, the normalization constraint is usually implemented as $\langle u| \mathbf{C}_0 |u\rangle = 1$, as this choice simplifies the optimization problem. Formally, tICA solves the optimization problem:

$$\begin{aligned} |\hat{u}_1\rangle =& \mathrm{argmax}\, \langle u| \mathbf{C}_\tau |u\rangle \\ & \text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1 \end{aligned} \tag{B10}$$

using the same procedure as for PCA, I can solve this with Lagrange multipliers, and show that

$$\mathbf{C}_\tau |\hat{u}_1\rangle = \lambda_1 \mathbf{C}_0 |\hat{u}_1\rangle \,, \tag{B11}$$

.

The generalized eigenvalue λ_i is equal to the autocorrelation of the projected coordinate $\langle \hat{u}_i|x(t)\rangle$ at lag time τ. Assuming that this coordinate approximates a single relaxation mode of the dynamics, its autocorrelation decays exponentially, yielding

$$\lambda_i \approx e^{-\tau/t_i}, \tag{B12}$$

where t_i is the corresponding relaxation timescale. In practice, the relation is approximate because a tIC may contain contributions from multiple dynamical modes.

Similarly as done with PCA, I can iterate this procedure and define the second tIC as the direction of maximal autocorrelation at lag time τ that is *uncorrelated* with the first:

$$\begin{aligned} |\hat{u}_2\rangle =& \mathrm{argmax}\, \langle u| \mathbf{C}_\tau |u\rangle \\ & \text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1 \\ & \text{and } \langle u| \mathbf{C}_0 |\hat{u}_1\rangle = 0 \end{aligned} \tag{B13}$$

the solution $|\hat{u}_1\rangle$ is again a solution of the same generalized eigenvalue problem as in Eq.(B10). generalized eigenvalue problem. Dimensionality reduction ($d < D$) is achieved by taking the components of the data $\mathbf{x}(t)$ along the first d tICs $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$\mathbf{z}(t) = (\langle x(t)|\hat{u}_i\rangle, \cdots, \langle x(t)|\hat{u}_d\rangle) \tag{B14}$$

By construction, in the reduced space components z_i have unit variance, are uncorrelated at zero lag, and are ordered by decreasing autocorrelation at lag time τ.

$$\mathbb{E}[|z\rangle \langle z|] = \mathbb{I} \tag{B15}$$

tICA as a diagonalization in the whitened coordinate space It is worth noting that unlike in PCA, the tIC vectors are generally not orthogonal under the standard Euclidean inner product. Instead, as I said, they satisfy the generalized orthogonality relation

$$\langle u_i| \mathbf{C}_0 |u_j\rangle = \delta_{ij}.$$

i.e. they corresponding components z_i and z_j are *uncorrelated* at zero lag. Nevertheless, the tICA procedure can be reformulated as a standard eigenvalue problem in a transformed coordinate system. The key idea is to first apply a whitening transformation, which rescales and rotates the original data so that all directions have unit variance and are mutually uncorrelated. Define the ZCA whitening transformation

$$|y\rangle = \mathbf{C}_0^{-1/2} |x\rangle$$

where $C_0^{-1/2}$ is symmetric and positive definite. By construction, the covariance matrix in the transformed space is the identity:

$$\mathbb{E}\left[|y\rangle \langle y|\right] = \mathbf{C}_0^{-1/2} \mathbf{C}_0 \mathbf{C}_0^{-1/2} = \mathbf{I}.$$

and time-lagged covariance matrix in the whitened space is

$$\hat{\mathbf{C}}_\tau = \mathbf{C}_0^{-1/2}\mathbf{C}_\tau\mathbf{C}_0^{-1/2}$$

$\hat{\mathbf{C}}_\tau$ is symmetric and can thus be diagonalized:,

$$\hat{\mathbf{C}}_\tau \left|v_i\right\rangle = \lambda_i \left|v_i\right\rangle, \qquad \langle v_i|v_j\rangle = \delta_{ij}.$$

If $|u_i\rangle$ solves the tICA problem (B10), then $|v_i\rangle = \mathbf{C}_0^{1/2}|u_i\rangle$ is an eigenvector of $\hat{\mathbf{C}}_\tau$ with the same eigenvalue.

Indeed, multiplying the generalized eigenvalue problem (B10) from the left by $\mathbf{C}_0^{-1/2}$ and defining $|v_i\rangle = \mathbf{C}_0^{1/2}|u_i\rangle$ yields

$$\hat{\mathbf{C}}_\tau \left|v_i\right\rangle = \lambda_i \left|v_i\right\rangle.$$

Consequently, the tICA components z_i correspond to the ortogonal projection of the whitened data $\mathbf{y}$ onto the ortonormal basis defined by $\{|v_i\rangle\}$:

$$z_i(t) = \langle x(t)|u_i\rangle = \langle x(t)|C_0^{-1/2}v_i\rangle = \langle y(t)|v_i\rangle.$$

Alternative generalized eigenvector normalizations for tICA

The construction presented above corresponds to the original formulation of tICA, in which the covariance matrix of the transformed data is the identity (B15). HoIver, alternative rescaling of the transformed coordinates have been used in the literature and are usually implemented in softwares to endow Euclidean distances in the reduced space with a more direct kinetic interpretation. These can be particularly useful when the tICA coordinates are subsequently clustered to construct a Markov state model, as I do in this work.

The most commonly used choice is the *kinetic map* in which each tIC component z_i is multiplied by its corresponding eigenvalue,

$$\tilde{z}_i(t) = \lambda_i z_i(t). \tag{B16}$$

With this normalization, the coordinates representing the sloIst processes "Iight more", and a unit step in those directions in the reduced space is assigned a larger distance. An alternative is the *commute map*, where each coordinate is rescaled according to its implied timescale,

$$\tilde{z}_i(t) = \sqrt{\frac{t_i}{2}}\, z_i(t), \qquad t_i = -\frac{\tau}{\ln \lambda_i}. \tag{B17}$$

With this normalization, Euclidean distances approximate commute distances, i.e. the expected time required to travel from one configuration to another and back. This is a variation on the same idea behind the kinetic map, being eigenvalues and timescales connected by (B12).

S2. Elements of Markov theory

Transition matrix and spectral decomposition. A Markov state model approximates the conformational dynamics as a memoryless jump process on the discrete microstates, encoded in the transition matrix $T(\tau)$ estimated at a fixed lag time τ. Following the convention used in DEEPTIME [**?**], T is taken to be row-stochastic,

$$\sum_j T_{ij} = 1, \qquad T_{ij} \geq 0, \tag{B18}$$

so that T_{ij} is the conditional probability of being in state j one lag step after being in state i. Probability distributions are represented as row vectors and evolve by right multiplication,

$$p^{(n)} = p^{(0)}\, T^{\,n}. \tag{B19}$$

The stationary distribution π is the left eigenvector with unit eigenvalue, $\pi T = \pi$ with $\sum_i \pi_i = 1$, and the estimator used here enforces detailed balance, $\pi_i T_{ij} = \pi_j T_{ji}$.

Assuming T is diagonalisable, it admits the spectral decomposition

$$T = \sum_{i=1}^{N} \lambda_i\, r_i\, l_i^{\mathsf{T}}, \tag{B20}$$

where the eigenvalues are ordered as $1 = \lambda_1 > \lambda_2 \geq \ldots$, and r_i, l_i are the right and left eigenvectors,

$$T r_i = \lambda_i r_i, \qquad l_i^{\mathsf{T}} T = \lambda_i l_i^{\mathsf{T}}, \tag{B21}$$

normalised to be bi-orthonormal, $l_i^{\mathsf{T}} r_j = \delta_{ij}$. The leading pair is fixed by $r_1 = \mathbf{1}$ (the constant vector, since T is row-stochastic) and $l_1 = \pi$. Substituting (B20) into (B19) and using bi-orthonormality, any initial distribution relaxes to equilibrium as a sum of exponentially decaying modes,

$$p^{(n)} = \sum_i c_i\, \lambda_i^{\,n}\, l_i^{\mathsf{T}}, \qquad c_i = p^{(0)} r_i. \tag{B22}$$

The dominant term ($\lambda_1 = 1$) is the stationary distribution π; because $|\lambda_i| < 1$ for $i \geq 2$, every other mode decays with a characteristic relaxation (implied) timescale

$$t_i = -\frac{\tau}{\ln \lambda_i}. \tag{B23}$$

Relaxation modes and probability flux. The left eigenvectors $\{l_i\}_{i\geq 2}$ describe the spatial structure of the relaxation processes: states whose components have opposite sign exchange probability on the associated timescale t_i, and the magnitude of each component measures how strongly that state participates in the process. The redistribution driven by these modes over a single lag step is obtained by expanding a distribution in the same basis, $p = \sum_i c_i l_i^{\mathsf{T}}$ with $c_i = p\, r_i$, and forming the difference

$$\Delta p = pT - p = \sum_{i\geq 2} c_i\,(\lambda_i - 1)\, l_i^{\mathsf{T}}, \tag{B24}$$

where the stationary term drops out because $\lambda_1 = 1$. Equation (B24) makes explicit that only the sub-unit eigenvalues carry probability flux, that the slowest modes ($\lambda_i \to 1$) produce the smallest per-step change and therefore dominate the long-time relaxation, and that the geometric pattern of each flux contribution is set by the corresponding left eigenvector l_i.

For a reversible chain the left and right eigenvectors are not independent but are related through the stationary distribution,

$$l_i = \Pi\, r_i, \qquad \Pi = \mathrm{diag}(\pi), \tag{B25}$$

that is, $l_i(x) = \pi_x\, r_i(x)$. The right eigenvectors are thus the equilibrium-population-weighted counterparts of the relaxation modes, and they are orthonormal in the π-weighted inner product,

$$\langle r_i, r_j\rangle_\pi := \sum_x \pi_x\, r_i(x)\, r_j(x) = \delta_{ij}. \tag{B26}$$

The sign structure of the dominant eigenvectors provides the geometric criterion used by PCCA++ [7, Ch. 2.5.1–2.5.2] to lump microstates into metastable macrostates. I use this to connect the slowest exchange modes of the Markov chain to the folded/unfolded thermodynamics of the protein, keeping in mind that spectral metastability and thermodynamic stability are related but not strictly identical notions.

Implied-timescale test and lag-time selection. Markovianity was assessed, and the lag time selected, through the implied-timescale (ITS) convergence test. The estimated timescales (B23) are variational lower bounds on the true timescales t_i^* of the underlying continuous-state process, $t_i(\tau) \leq t_i^*$, with the gap shrinking as the discretisation is refined or as τ is increased. In practice $t_i(\tau)$ is plotted against τ and one looks for a range over which the timescales are approximately constant; the lag time is taken as the smallest τ (counted in frames, $\tau = 1, 2, 3, \ldots$) inside this range, which minimises the loss of temporal resolution while keeping the process of interest approximately Markovian [7, Ch. 4.7]. Because the number of microstates and the lag time jointly control resolution and statistical accuracy, the two were chosen together (see main text, Sec. III A).